Resampling methods: Cross-validation

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March 27, 2025

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Cross-validation

Validation

Thinking about the true loss function is important

Most of the regression methods we've studied aim to minimize the RSS, while classification methods aim to minimize the 0-1 loss.

In classification, we often care about certain kinds of error more than others; i.e. the natural loss function is not the 0-1 loss.

Even if we use a method which minimizes a certain kind of training error, we can tune it to optimize our true loss function.

Example: in the default study we could find the threshold that brings the False negative rate below an acceptable level.

Cross-validation

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- the number of variables to include in forward or backward selection or best subset.
- the tuning parameter in shrinkage regression (Ridge, Lasso)
- the order of polynomial in a polynomial regression.

Validation Set Approach

Goal: Estimate the test error for a supervised learning method.

Strategy:

1. Split the data in two parts.
2. Train the method on the first part.
3. Compute the error on the second part.

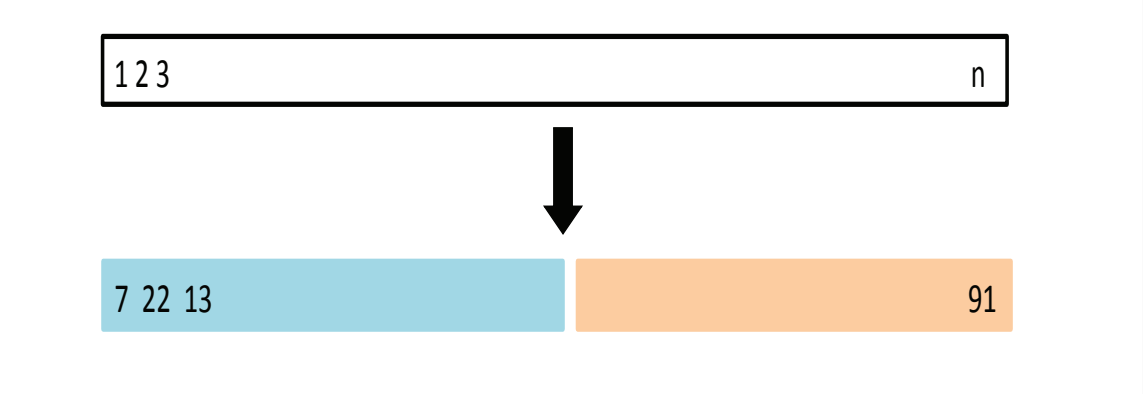

Figure 1: A schematic display of the validation set approach. A set of n observations are randomly split into a training set (shown in blue, containing observations 7, 22, and 13, among others) and a validation set (shown in beige, and containing observation 91, among others). The statistical learning method is fit on the training set, and its performance is evaluated on the validation set.

Polynomial Regression

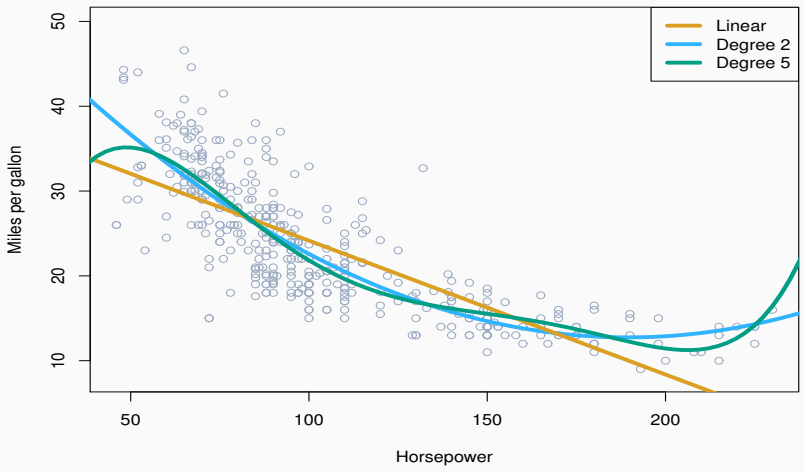

Figure 2: The Auto data set. For a number of cars, mpg and horsepower are shown.

- A scatterplot between a predictor and the response may reveal a non-linear relationship.
- Solution: include polynomial terms in the model.

$$\texttt{MPG} = \beta_0 + \beta_1 \times \texttt{horsepower} + \beta_2 \times \texttt{horsepower}^2 + \epsilon$$

Order of polynomial

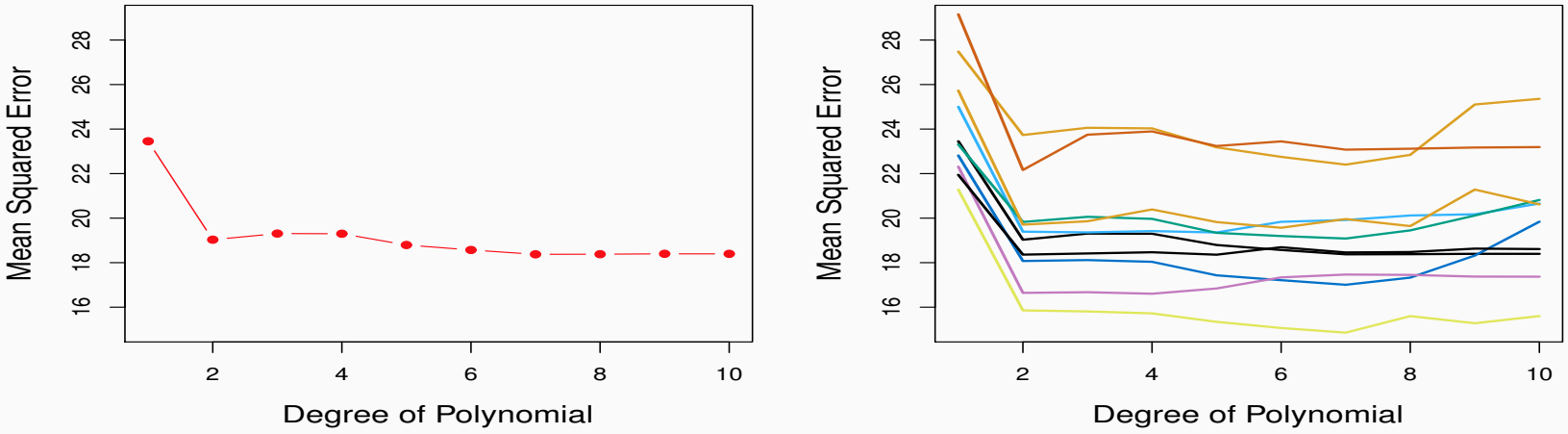

Figure 3: The validation set approach was used on the Auto data set in order to estimate the test error that results from predicting mpg using polynomial functions of horsepower. Left: Validation error estimates for a single split into training and validation data sets. Right: The validation method was repeated ten times, each time using a different random split of the observations into a training set and a validation set. This illustrates the variability in the estimated test MSE that results from this approach.

Issue with Validation Set

1. Every split yields a different estimate of the error.
2. Only a subset of points is used to evaluate the model.

Cross-validation is a refinement that takes care of this issue.

Leave-one-out Cross Validation / LOOCV

- For every $i = 1, \ldots, n$:
  - train the model on every point except i
  - compute the test error on the held out point.
- Average the test errors.

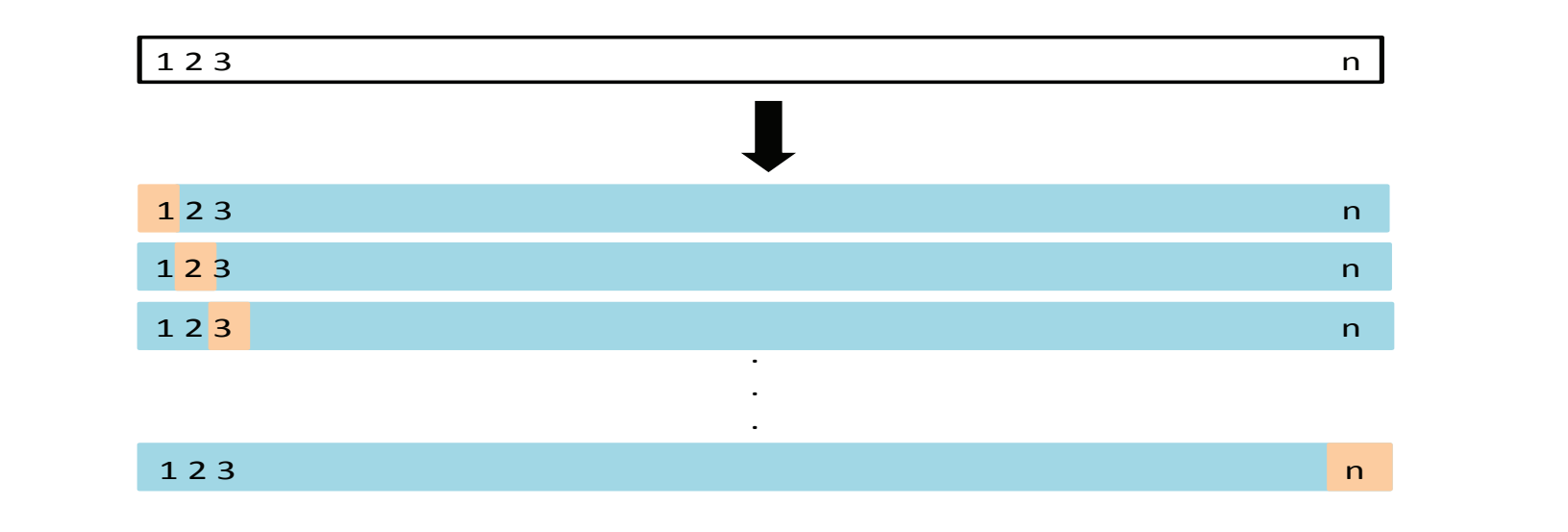

Figure 4: A schematic display of LOOCV. A set of n data points is repeatedly split into a training set (shown in blue) containing all but one observation, and a validation set that contains only that observation (shown in beige). The test error is then estimated by averaging the n resulting MSE's.

LOOCV: Regression

- For every $i = 1, \ldots, n$:
  - train the model on every point except i
  - compute the test error on the held out point.
- Average the test errors:

$$CV_n = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i^{-i})^2$$

- Regression: Prediction for i^{th} sample without using the ith sample.

LOOCV: Classification

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  - train the model on every point except i
  - compute the test error on the held out point.
- Average the test errors:

$$CV_n = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}(y_i \neq \hat{y}_i^{-i})$$

- Classification problem: predicted label for i^{th} sample without using the ith sample.

Pros and Cons

- Pros:
  1. Less Bias than Validation Set Approach : repeatedly fit the method using $n - 1$ observations: almost as many as entire data set. In validation set, training set is typically half the size of entire data.
  2. No randomness in training / validation set splits: performing LOOCV multiple times will yield the same result.

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- Luckily, for linear regression there is a shortcut:

$$CV_n = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_{ii}}\right)^2$$

- where h_{ii} is the leverage statistics.

k-fold CV

An alternative to LOOCV is k-fold CV.

Strategy:

1. Split the data into k subsets or folds.
2. For every $i = 1, 2, \ldots, k$
    3. train the model on every fold except the ith fold,
    4. Compute the test error on the ith fold.
5. Average the test errors:

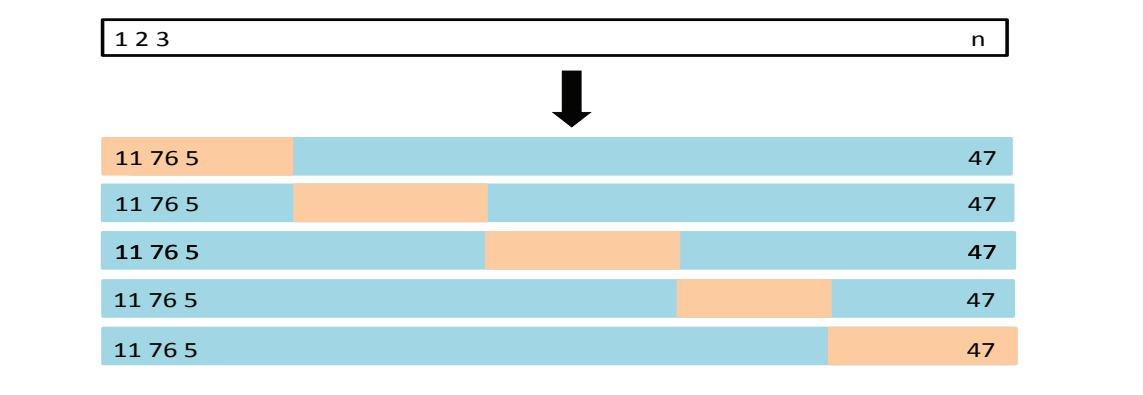

Figure 5: A schematic display of 5-fold CV. A set of n observations is randomly split into five non-overlapping groups. Each of these fifths acts as a validation set (shown in beige), and the remainder as a training set (shown in blue). The test error is estimated by averaging the five resulting MSE estimates

k-fold vs. LOOCV

- k-fold CV depends on the chosen split.
- In k-fold CV, we train the model on less data than what is available. This introduces bias into the estimates of test error.
- In LOOCV, the training samples highly resemble each other. This increases the variance of the test error estimate.

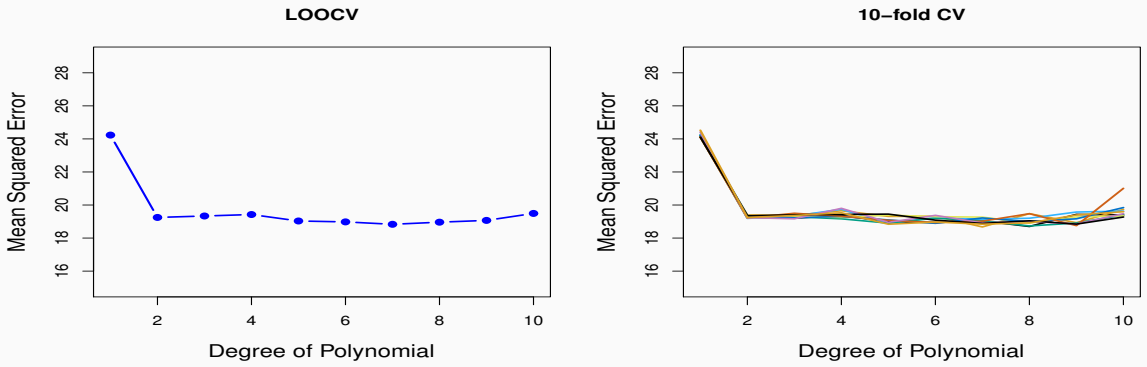

Figure 6: Left: The LOOCV error curve. Right: 10-fold CV was run nine separate times, each with a different random split of the data into ten parts. The figure shows the nine slightly different CV error curves.

Some Remarks/Comments

- K-fold CV depends on the chosen split (somewhat).
- In K-fold CV, we train the model on less data than what is available to LOOCV. This introduces bias into the estimates of test error.
- In LOOCV, the training samples highly resemble each other. This increases the variance of the test error estimate.
- n-fold CV is equivalent LOOCV (where n is the sample size).

Choosing an optimal model

We'll quickly recall these three simulated case studies we've seen before:

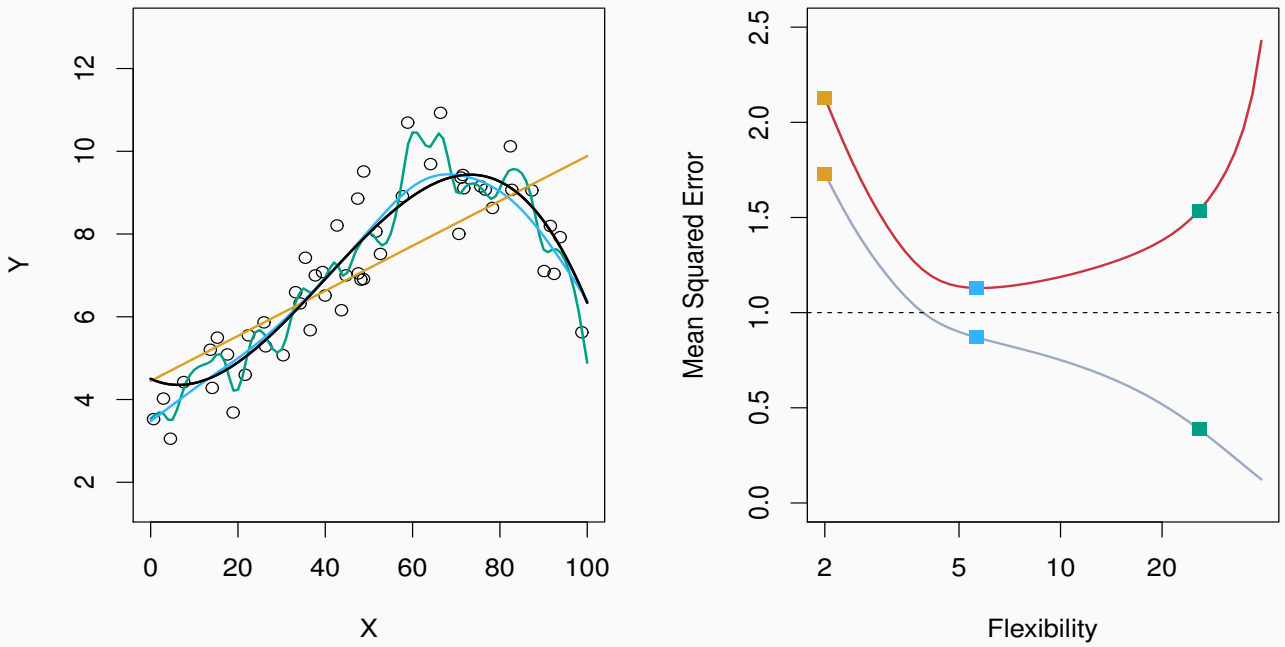

Figure 7: Data simulated from f, shown in black. Three estimates of f are shown: the linear regression line (orange curve), and two smoothing spline fits (blue and green curves)

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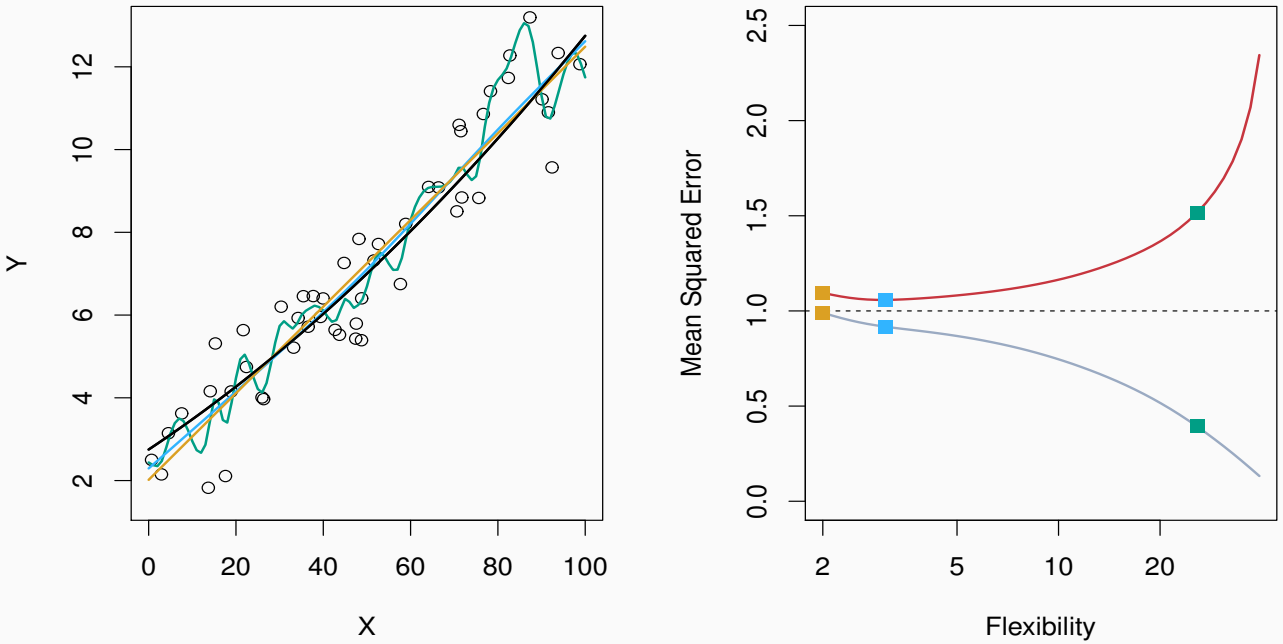

Figure 8: Same as before, f is almost-linear

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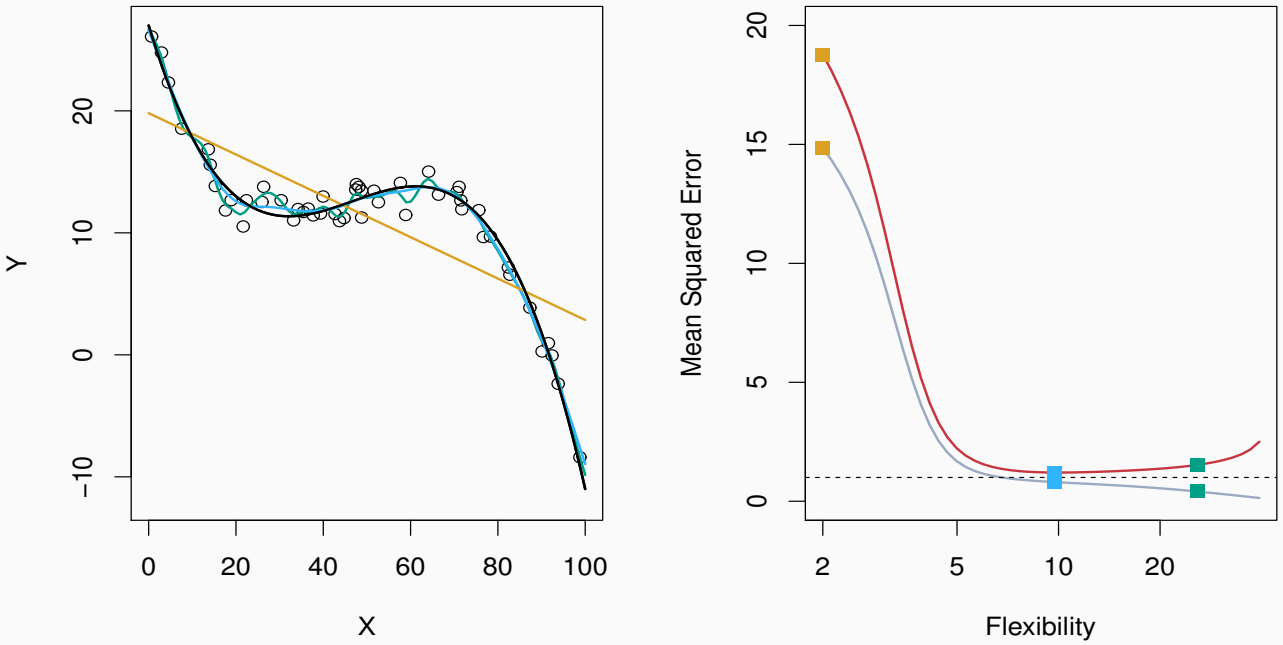


Figure 9: Same as before, f is highly non-linear

Choosing an optimal model

True and estimated test MSE for the three simulated data sets. Even if the error estimates are off, choosing the model with the minimum cross validation error often leads to a method with near minimum test error.

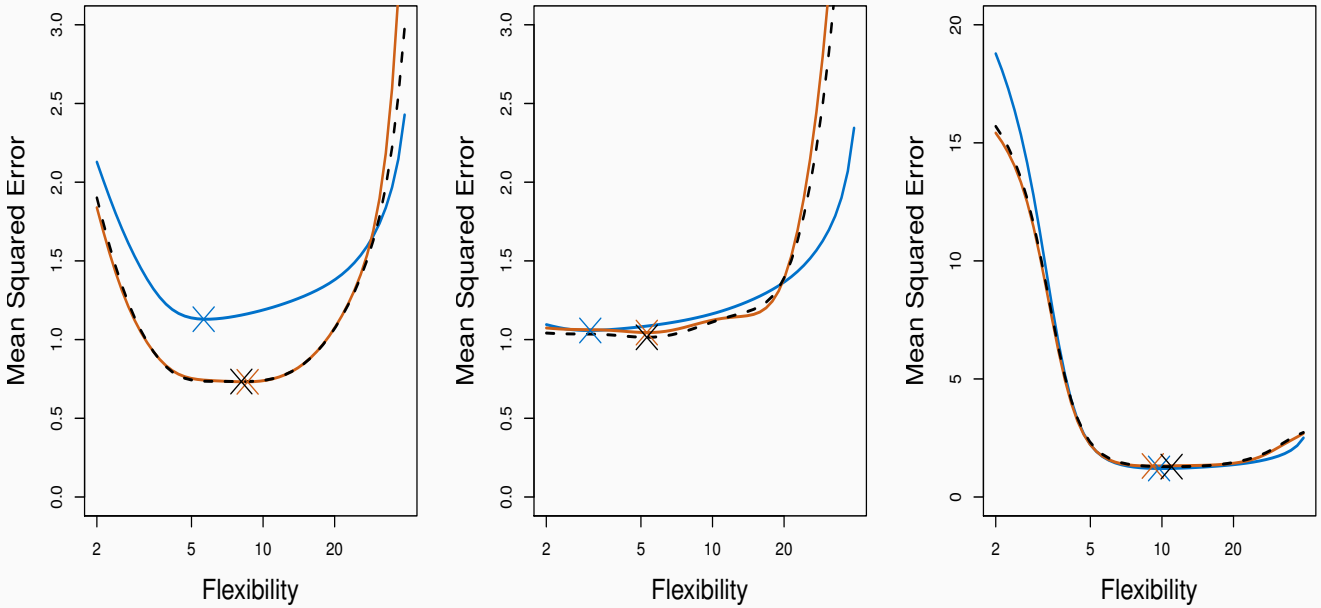
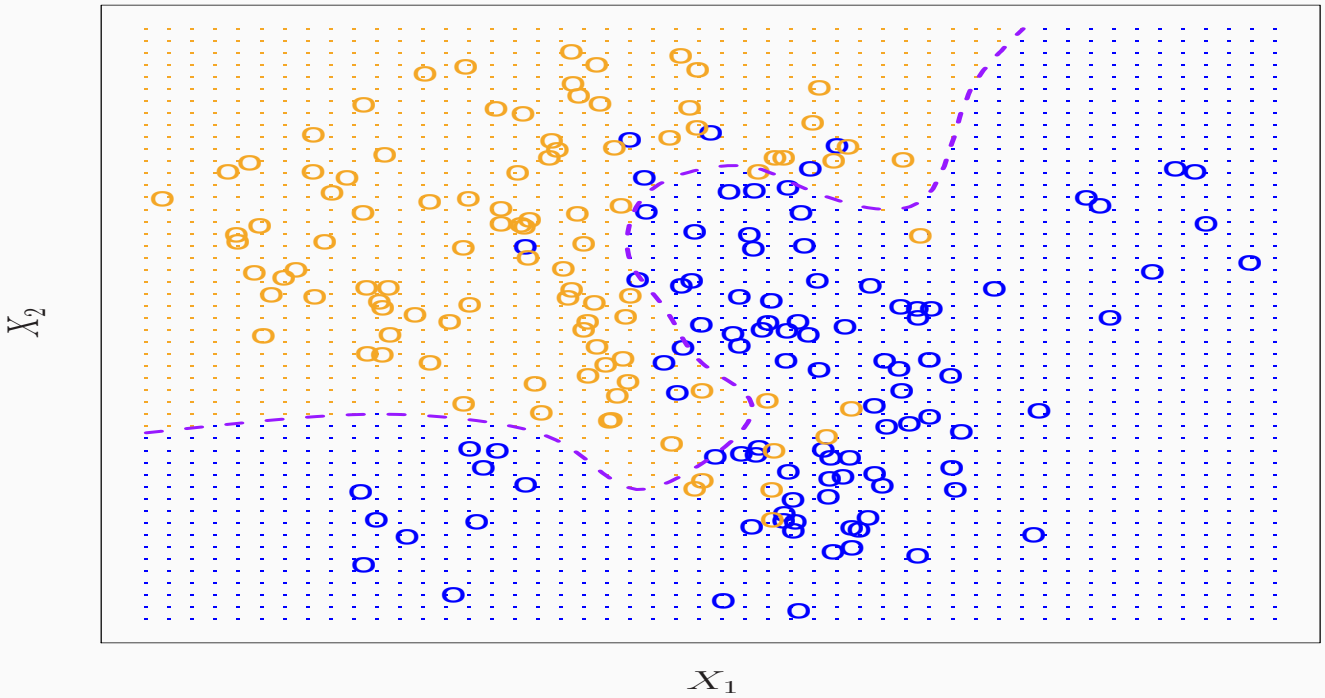


Figure 10: The true test MSE is shown in blue, the LOOCV estimate is shown as a black dashed line, and the 10-fold CV estimate is shown in orange. The crosses indicate the minimum of each of the MSE curves.

Classification

In a classification problem, things look similar. We'll look at this earlier example.

Since this is simulated data, we can compute the true test error rate, which takes a value of 0.201 and so is substantially larger than the Bayes error rate of 0.133.



Classification

In a classification problem, things look similar. The test error rates for the four logistic regression fits are respectively 0.201, 0.197, 0.160, and 0.162, while the Bayes error rate is 0.133.

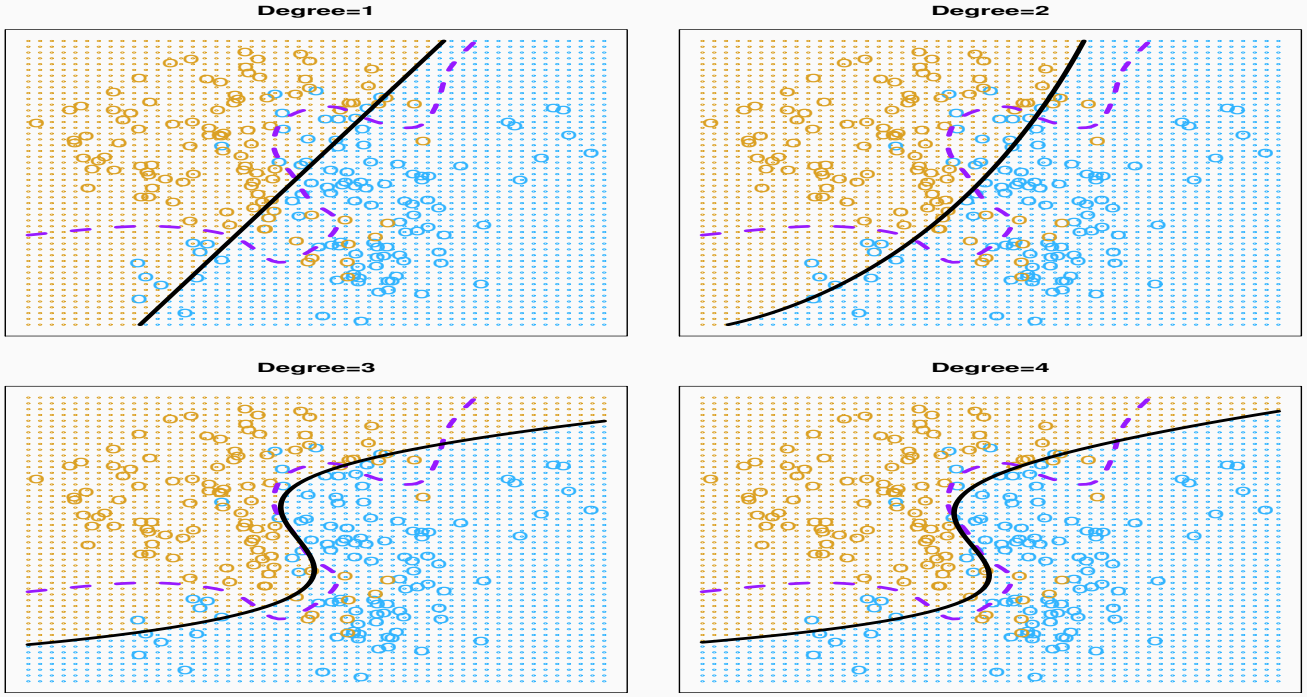


Figure 11: Logistic regression fits on the two-dimensional classification data displayed before. The Bayes decision boundary is represented using a purple dashed line. Estimated decision boundaries from linear, quadratic, cubic and quartic (degrees 1–4) logistic regressions are displayed in black.

Extended Logistic models

Clearly logistic regression does not have enough flexibility to model the Bayes decision boundary in this setting.

We can easily extend logistic regression to obtain a non-linear decision boundary by using polynomial functions of the predictors.

For example, we can fit a quadratic logistic regression model, given by

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_2 + \beta_4 X_2^2$$

- Logistic: Cubic model has best test error, Quartic has best CV. Curves look similar.
- KNN: Though the cross-validation error curve slightly underestimates the test error rate, it takes on a minimum very close to the best value for K.

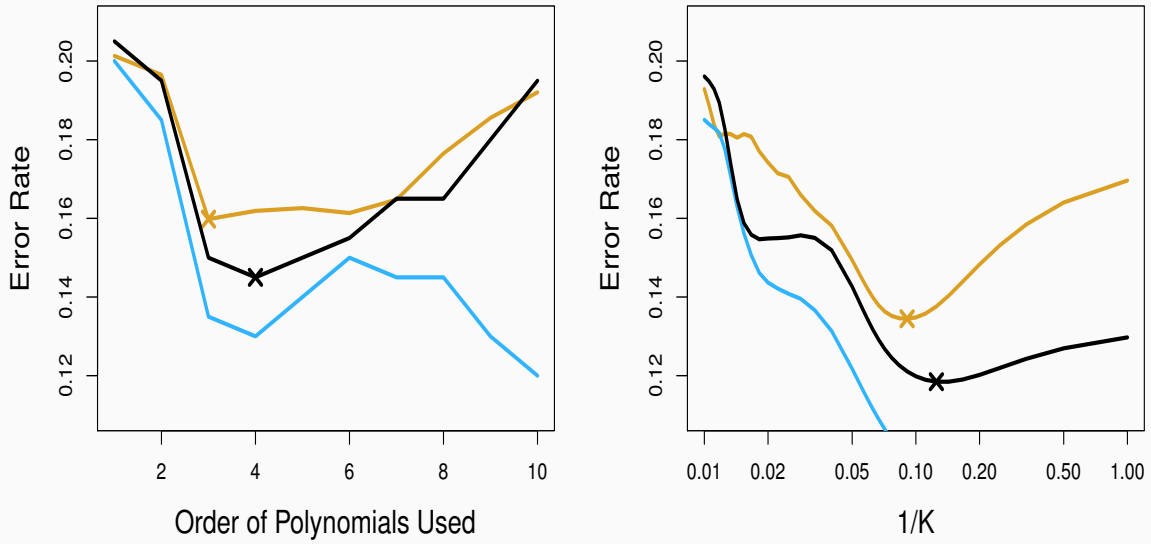


Figure 12: Test error (brown), training error (blue), and 10-fold CV error (black) on the two-dimensional classification data displayed in last Figure. Left: Logistic, Right: KNN.

1-SE rule of thumb

- A number of models with $10 \leq p \leq 15$ have almost the same CV error. The vertical bars represent 1 standard error in the test error from the 10 folds.
- Choose the simplest model whose CV error is no more than one standard error above the model with the lowest CV error.

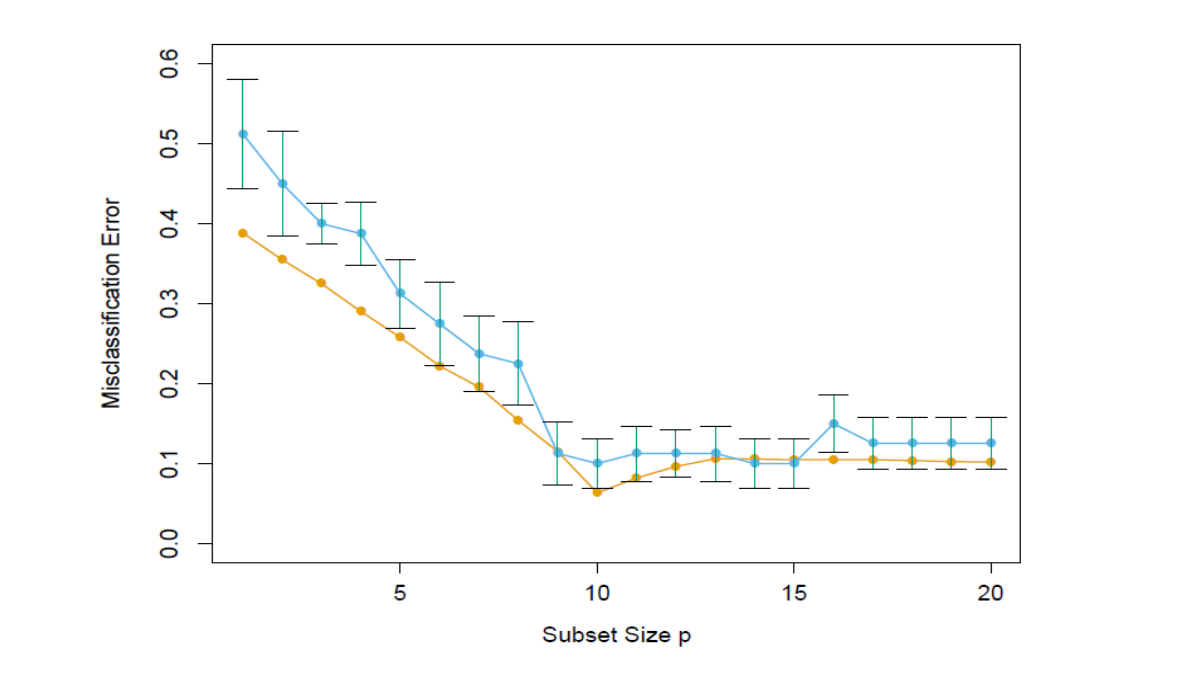


Figure 13: Prediction error (orange) and tenfold cross-validation curve (blue) estimated from a single training set.

R examples and packages

https://dattahub.github.io/data-analytics/crossvalidation.html